

Week 8 Report: Model Evaluation

Project: Analysing the Role of Big Data Analytics for Cyber Threat Detection and Prevention in Banking Organisations of Ireland

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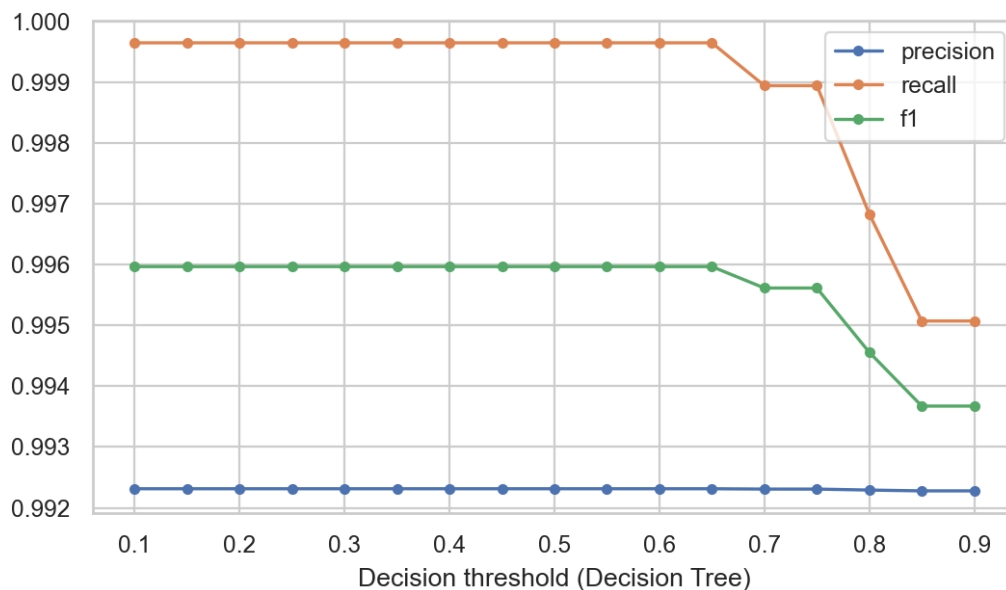
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What I did this week

This week I checked what drives the results. First, imbalance handling makes no difference here: Random Forest reaches F1 0.996 with no re-balancing, with class weights and with SMOTE. That makes sense because 42% of rows are suspicious, so the data is not strongly imbalanced.

Second, the ablation shows where the signal is. Without the description and channel, F1 falls to 0.800. With amount and balance only it is 0.745. The models depend heavily on which merchant the transaction is with.

Feature set	F1	ROC-AUC
All features	0.996	0.997
Without description/channel	0.800	0.890
Without date features	0.996	0.998
Description/channel only	0.839	0.862
Amount & balance only	0.745	0.772



Precision and recall of the best model at different thresholds.

Next week

Run the robustness tests: a time-based split and a test on merchants the model has never seen.